

Business Requirements Document (BRD)

Project Title

Internship Project: Web Application Development using .NET 10

Document Version

- Version: 2.0
 - Date: 2026
 - Prepared For: .NET Interns
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1. Introduction

1.1 Purpose of the Document

This Business Requirements Document (BRD) defines the scope, objectives, functional and non-functional requirements for a **.NET 10-based web application** to be developed by interns. The document acts as a bridge between business needs and technical implementation, helping interns gain exposure to modern, enterprise-grade .NET development.

1.2 Project Overview

The project involves building a **secure, role-based web application** using **ASP.NET Core on .NET 10**. Interns will learn modern development concepts such as clean architecture, RESTful APIs, authentication and authorization, database access, logging, and error handling using the latest .NET platform.

1.3 Target Audience

- .NET Interns
 - Technical Mentors / Trainers
 - Project Supervisors
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2. Business Objectives

- Provide hands-on experience with **.NET 10 and ASP.NET Core**
 - Familiarize interns with **modern software development lifecycle (SDLC)**
 - Introduce **clean architecture and separation of concerns**
 - Enable practical understanding of **security, data access, and APIs**
 - Improve coding standards, debugging skills, and documentation practices
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3. Scope of the Project

3.1 In-Scope

- Development of a web-based application using .NET 10
- Role-based authentication and authorization
- CRUD operations using Entity Framework Core
- RESTful API development (Controller-based or Minimal APIs)
- Server-side UI using Razor Pages or MVC
- Logging, validation, and global exception handling
- Local deployment using Kestrel / IIS Express

3.2 Out of Scope

- Mobile or desktop application development
 - Third-party payment gateway integration
 - Advanced cloud-native deployment
 - High-volume performance or stress testing
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4. Proposed Application Description

4.1 Application Name (Sample)

Intern Management System (IMS)

4.2 Application Description

The Intern Management System (IMS) is a web application that allows administrators to manage intern profiles, assign tasks, and monitor progress. Intern users can log in to view assigned tasks, update task status, and submit periodic progress reports.

5. Stakeholders

Role	Responsibility
Intern	Design and develop assigned features
Mentor	Code review, guidance, and technical support
Project Supervisor	Monitor progress and approve deliverables

6. User Roles

6.1 Admin

- Secure login
- Create, update, and deactivate users
- Assign and manage intern tasks
- View and review progress reports

6.2 Intern

- Secure login
 - View assigned tasks
 - Update task status
 - Submit daily or weekly progress reports
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7. Functional Requirements

7.1 Authentication & Authorization

- Login using username and password
- Role-based authorization (Admin, Intern)
- Secure session or token-based authentication

7.2 User Management

- Admin can create, update, and deactivate users
- Passwords must be securely hashed

7.3 Task Management

- Admin can create, assign, and update tasks
- Intern can update task progress
- Task statuses: Pending, In Progress, Completed

7.4 Progress Reporting

- Intern can submit progress reports
- Admin can view reports by intern and date range

7.5 Logging & Audit

- Application logs user actions
 - System errors and exceptions are logged
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8. Non-Functional Requirements

8.1 Performance

- Average response time should be under 3 seconds for standard operations

8.2 Security

- HTTPS enabled
- Secure password storage
- Authorization enforced at controller/API level

8.3 Usability

- Clean and simple user interface
- Proper validation and user-friendly error messages

8.4 Maintainability

- Follow clean architecture principles
 - Consistent coding standards
 - Modular and well-documented codebase
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9. Technical Requirements

9.1 Technology Stack

- Framework: .NET 10
- Language: C# (latest supported version)
- Web Framework: ASP.NET Core (MVC / Razor Pages / Minimal APIs)
- Database: SQL Server
- ORM: Entity Framework Core
- Authentication: ASP.NET Core Identity
- Logging: ILogger with structured logging

9.2 Architecture

- Presentation Layer (UI / API)
 - Application / Business Logic Layer
 - Data Access Layer
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10. Database Requirements (High Level)

Key Entities

- Users
 - Roles
 - Intern Profiles
 - Tasks
 - Progress Reports
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11. Assumptions & Constraints

Assumptions

- Interns have basic knowledge of C# and OOP
- Development will be done in a local environment

Constraints

- Project timeline limited to internship duration
 - Use of only built-in or approved open-source libraries
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12. Deliverables

- Complete source code repository
 - Database schema or migration scripts
 - Technical and user documentation
 - Final demo or presentation
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13. Acceptance Criteria

- All functional requirements are implemented
 - Application builds and runs successfully on .NET 10
 - Role-based access control works correctly
 - CRUD operations perform without errors
 - Code follows clean architecture and best practices
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14. Future Enhancements (Optional)

- Email or notification system
- Admin dashboard with charts and metrics

- Export reports to Excel or PDF
- Deployment to Azure App Service

15. Approval

Name	Role	Signature	Date

End of Document